Technical specification –

Automatic Retractable Fabric Roof System,

Scope of Work

Design, engineering, supply, installation, testing, and commissioning of an Automatic Retractable Fabric Roof System, with PVC fabric, structural framework, motorized drive mechanism, control system, and all required accessories for smooth, safe, and reliable operation.

1. Fabric Specification

- **Fabric Type** **:** Imported architectural-grade retractable roof fabric
- **Material** **:** PVC-coated polyester
- **Fabric Weight** **:** Minimum 750–950 GSM
- **Tensile Strength** **:** High tensile and tear resistance
- **UV Protection** **:** UV-stabilized, anti-fading treatment
- **Waterproofing** **:** 100% waterproof with heat-welded seams
- **Fire Retardancy** **:** Fire-retardant as per international standards (DIN 4102 B1 / EN 13501 or equivalent)

- **Weather Resistance** **:** Resistant to rain, dust, pollution, mildew, and fungus
- **Standard Colors** **:** White, Light Grey
    - **Optional Colors** **:** Available on request at additional cost

2. Structural Framework

- **Main Structure** **:** High-strength aluminium alloy / hot-dip galvanized steel profiles
- **Surface Treatment** **:** Powder-coated finish for corrosion resistance
- **Support System** **: Fixed** beams, movable carriages, and guide rails

- **Fasteners & Accessories:**
    - Stainless steel (SS 304 / SS 316) fasteners and corrosion-resistant hardware

3. Retractable Mechanism

- **System Type:** Sliding / folding retractable roof system (as per design)
- **Movement System: T**rack-guided synchronized movement with rollers / carriages

4. Motor & Drive System

- **Operation Mode:** Fully motorized
- **Motor Type** **:** Heavy-duty tubular / geared motor suitable for retractable roof applications
- **Power Supply** **:** 230V AC, 50 Hz
- **Protection Class :** IP-rated motor suitable for outdoor installation
- **Limit Control:** Adjustable electronic limit switches for open and close positions

5. Control System

- Wall-mounted push button switch
- Handheld remote control (standard)
- Manual override facility in case of power failure
- **Optional Automation (if required):**
    - Rain sensor
    - Integration with BMS / home automation system

6. Drainage & Weather Management

- Engineered roof slope to facilitate rainwater drainage
- Integrated gutter or drainage channels to avoid water stagnation
- Fabric designed to withstand normal wind loads; retraction recommended during high wind conditions

7. Electrical & Safety Features

- Overload protection for motor
- Emergency stop function
- Proper earthing and electrical safety compliance
- All electrical components suitable for outdoor conditions

8. Warranty

- **Fabric:** As per manufacturer's standard warranty 3 years
- **Motor & Electrical Components:** Minimum 12 months
- **Structural Frame:** Minimum 10 Years against manufacturing defects

9. Notes

- Standard fabric colors: **White and Light Grey**
- **Any other color or custom fabric finish will attract additional cost**
- Final system design subject to site survey, structural feasibility, and engineering approval